

Notes on Integrability from Gauge theory

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Chapter 1

The Yang-Baxter Equation

1.1 Introduction

The plan of this lecture is to introduce you to the Yang-Baxter equation [1] and its role in integrable systems through some historical examples. The three examples we will look at are: (a) the six-vertex model [2], (b) Hamiltonian mechanics [3], and (c) two-dimensional quantum field theories [4]. I will try to illustrate how the Yang-Baxter equation makes an appearance in each of the examples. The point I want to drive home through this lecture is that the Yang-Baxter equation is ubiquitous whenever we talk of integrability; although in some cases we will have to work extra hard to extract it.

The hero of our story is the R -matrix which is a linear operator on two copies of a finite-dimensional vector space V , i.e.,

$$R : V \otimes V \rightarrow V \otimes V.$$

The R -matrix parametrically depends on a complex number $z \in \mathbb{C}$, called the *spectral parameter*, and it satisfies the Yang-Baxter equation:

$$R_{12}(z_1 - z_2)R_{13}(z_1 - z_3)R_{23}(z_2 - z_3) = R_{23}(z_2 - z_3)R_{13}(z_1 - z_3)R_{12}(z_1 - z_2). \quad (1.1.1)$$

This is an equality between two linear operators acting on $V \otimes V \otimes V$ with R_{ij} acting as R on the i^{th} and j^{th} copy of V and as identity on the remaining copy. For example,

$$R_{12}(V \otimes V \otimes V) = R(V \otimes V) \otimes V,$$

i.e., R_{12} acts as R on the first two copies of V and as the identity on the third copy of V .

The Yang-Baxter equation is graphically represented in Fig. 1.1. We will make use of this picture when discussing the examples below. There is a precise meaning to this figure when we talk about scattering in $(1+1)$ -dimensional quantum field theories. As we will see there, the role of the R -matrix is played by the S -matrix of the QFT, and the Yang-Baxter equation is a consistency condition for an integrable QFT. The role of the spectral parameter will be played by the rapidity of the incoming and outgoing states.

How constraining is the Yang-Baxter equation? A simple counting suggests that there are $\sim d^4$ components of R while the Yang-Baxter equation is a set of $\sim d^6$ equations, where d is the dimension of the vector space V . Therefore, the Yang-Baxter equation is very constraining, and it seems miraculous that there are solutions. However, as we will

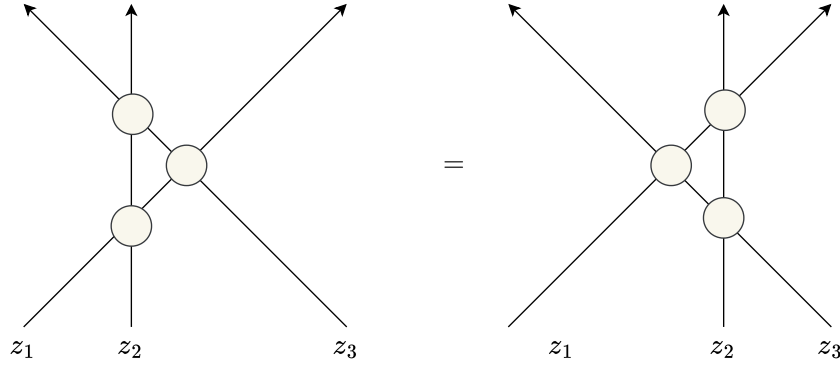


Figure 1.1: Graphical representation of the Yang-Baxter equation.

see in the examples below, there is a rich variety of non-trivial R -matrices that solve the Yang-Baxter equation.

The first example I'll introduce is the **six-vertex model** [2] which is a special case of Baxter's eight-vertex model [5]. But before we do that, I have the following exercise for you to work out.

Exercise 1.1. Consider an R -matrix of the form

$$R(z) = a + \frac{b}{z} \mathbb{P}$$

where $\mathbb{P}$ is the flip operator acting on $V \otimes V$ defined on a basis as

$$\mathbb{P} |i_1, i_2\rangle = |i_2, i_1\rangle.$$

Show that for any value of $a, b \in \mathbb{C}$, the above R -matrix satisfies the Yang-Baxter equation. In fact, this was the solution that appeared in Yang's original paper [6].

1.2 Six-vertex model

Vertex models are classical statistical mechanics models defined on a $N \times N$ square grid with periodic boundary conditions. The edges are oriented such that each vertex satisfies the *ice rule* [7]: number of incoming edges equals the number of outgoing edges. This leaves six types of vertices shown in Fig. 1.2, and hence the name six-vertex model.

Each of these vertices is assigned a weight $w_1, \dots, w_6$ and the partition function of the model is defined as

$$\mathcal{Z} = \sum \prod_{\text{edges vertices}} w_i \quad (1.2.1)$$

where the sum is over all allowed configurations subject to the ice rule. The meaning of the above equation is as follows. First, we assign an orientation to each edge of the lattice subject to the ice rule. The Boltzmann factor of this particular configuration of edge orientations is a product of the weight of each vertex. The partition function is then the sum over all allowed configurations.

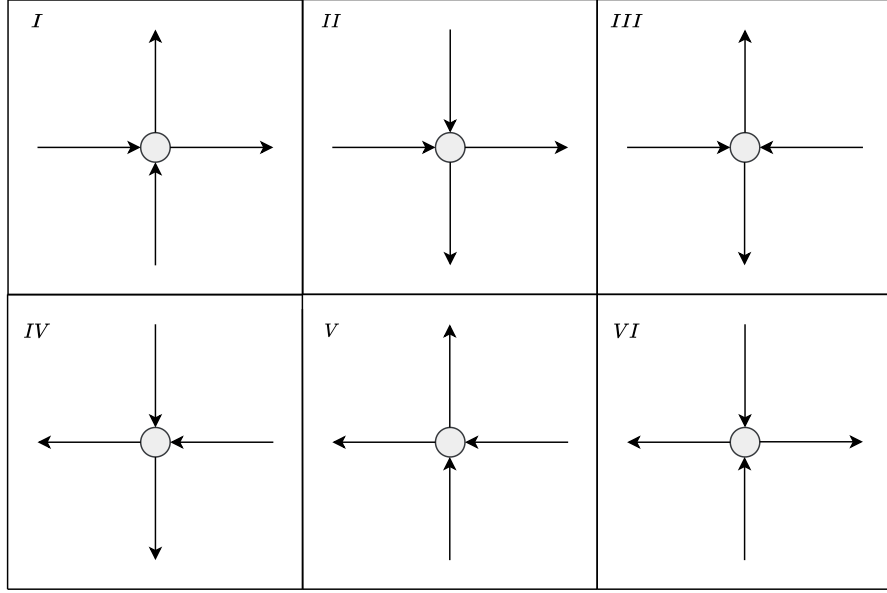


Figure 1.2: Six allowed vertices according to the ice rule.

1.2.1 The R -matrix

If we assume the model is invariant under a parity transformation, then we get three independent weights $a = w_1 = w_4$, $b = w_2 = w_5$, and $c = w_3 = w_6$. These weights can be arranged into a matrix defined as

$$R_{\alpha\beta}^{\gamma\delta} = \begin{array}{c} \delta \\ | \\ \alpha \text{ --- } \bigcirc \text{ --- } \gamma \\ | \\ \beta \end{array} \quad (1.2.2)$$

with the convention that $\alpha, \beta = +$ if the legs are incoming and $-$ if outgoing, and $\gamma, \delta = +$ if outgoing and $-$ if incoming. With this choice, we get the following non-zero elements of R :

$$R = \begin{pmatrix} a & & & \\ & b & c & \\ & c & b & \\ & & & a \end{pmatrix}. \quad (1.2.3)$$

Baxter [8] introduces a spectral parameter $u \in \mathbb{C}$ and parametrizes the weights as

$$a = \sin(\eta + u), \quad b = \sin(u), \quad c = \sin(\eta), \quad (1.2.4)$$

with η playing the role of a coupling constant. The significance of the above parametrization is highlighted via the following exercise.

Exercise 1.2. Show that the above parametrization of a , b , and c leaves

$$\Delta = \frac{a^2 + b^2 - c^2}{2ab}$$

invariant as η is held fixed and u is varied. In fact, you can easily show that $\Delta = \cos \eta$. Now, think of R as a function of u keeping η (alternatively Δ) fixed, acting on two copies $V \otimes V$ of a two-dimensional vector space V . With this interpretation of $R(u)$, show that it satisfies the Yang-Baxter equation! You may want to use *Mathematica* for this. In other words, you have to show that

$$R_{12}(u_1 - u_2)R_{13}(u_1 - u_3)R_{23}(u_2 - u_3) = R_{23}(u_2 - u_3)R_{13}(u_1 - u_3)R_{12}(u_1 - u_2)$$

with the interpretation of R_{ij} as in the Yang-Baxter equation.

At this point, the fact that $R(u)$ satisfies the Yang-Baxter equation might seem like a curiosity with no real consequence. Our choice of parametrization is rather arbitrary and there's no physical meaning to the variables η and u .

But the fact that $R(u)$ with this choice of parametrization satisfies the Yang-Baxter equation is crucial for the integrability of the six-vertex model. I'll not go into the full detail of Baxter's solution, but highlight how the R -matrix helps with the problem.

1.2.2 The Transfer matrix

To do this, we will need to introduce the *transfer matrix* which I'll graphically depict in Fig. 1.3 where the horizontal legs are contracted (shown using a slanted double line). In terms of an equation, it is given by

$$T(u)_{\alpha_1 \dots \alpha_N}^{\beta_1 \dots \beta_N} = \sum_{\mu_1, \dots, \mu_N} R(u)_{\mu_1 \alpha_1}^{\mu_2 \beta_1} R(u)_{\mu_2 \alpha_2}^{\mu_3 \beta_2} \dots R(u)_{\mu_N \alpha_N}^{\mu_1 \beta_N}. \quad (1.2.5)$$

The partition function of the six-vertex model on an $N \times N$ square lattice is

$$\mathcal{Z} = \text{Tr} (T(u)^N). \quad (1.2.6)$$

In the large N (thermodynamic) limit, the maximum eigenvalue of $T(u)$ will dominate the partition function.

I will now argue that the transfer matrices corresponding to different spectral parameters commute, i.e.,

$$[T(u), T(v)] = 0, \quad \forall u, v \in \mathbb{C}. \quad (1.2.7)$$

We will use the Yang-Baxter equation to derive this key result. For more details on how this result helps solve the six-vertex model, see [8].

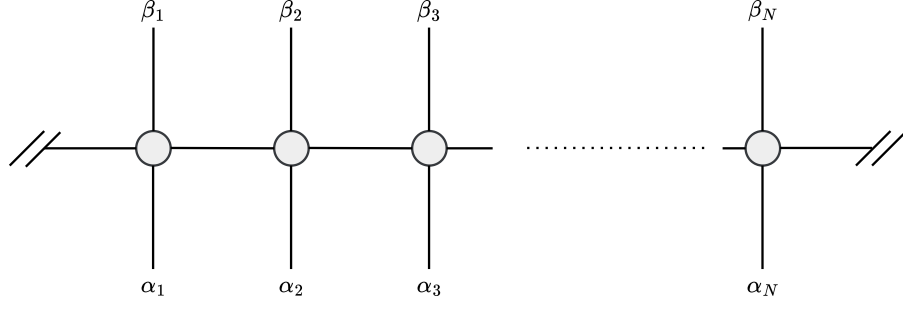
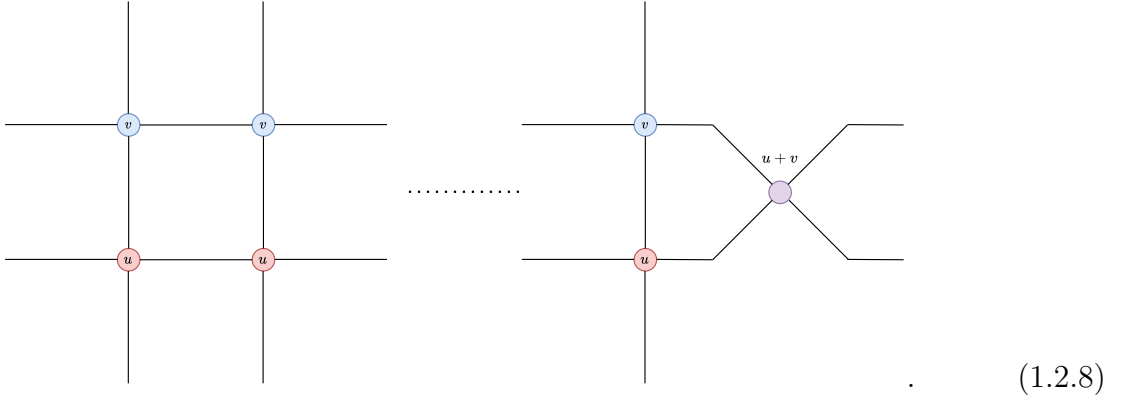


Figure 1.3: Graphical representation of the transfer matrix.

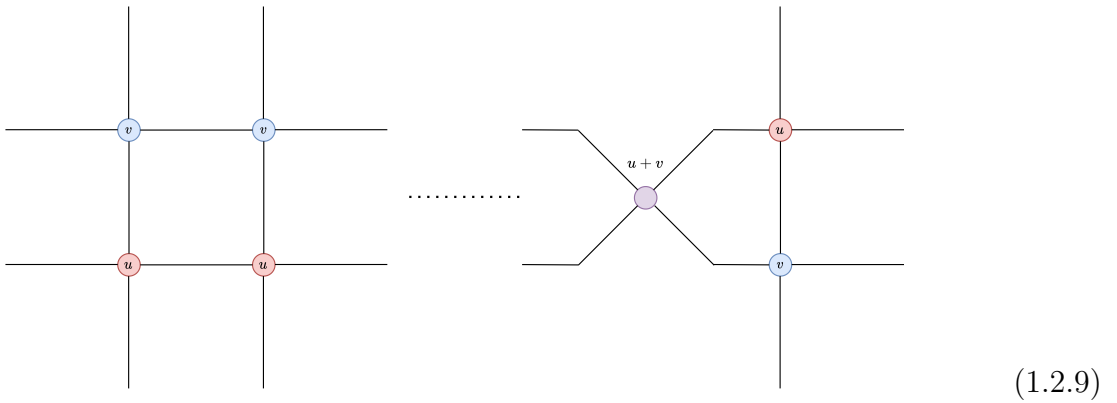
1.2.3 Commuting transfer matrices

To see that the transfer matrices commute, consider the tensor network below:



The red vertices represent $R(u)$, the blue vertices represent $R(v)$, and the purple vertex on the right is $R(u+v)$.

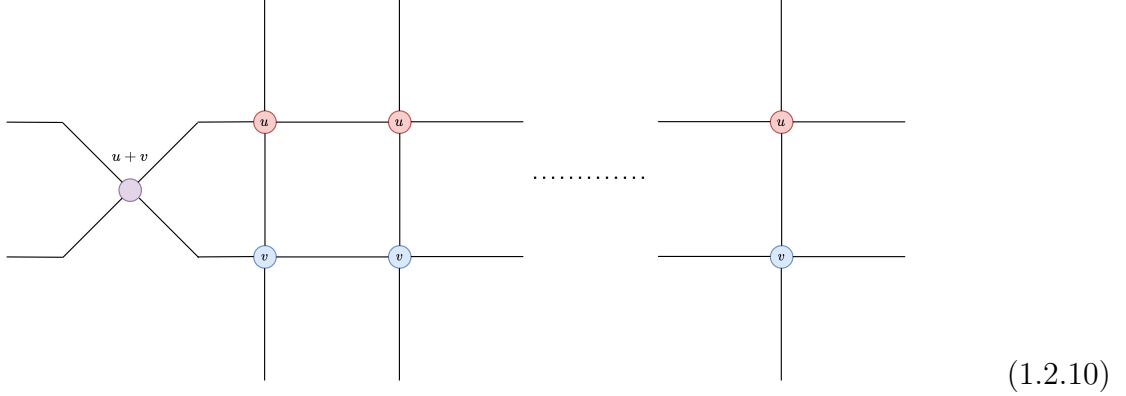
From the Yang-Baxter equation, we can move the purple vertex one step to the left while interchanging the red and blue vertices. This is shown in the figure below:



This is a valid move by virtue of the Yang-Baxter equation.

Proceeding this way, we can move the purple vertex all the way to the left of the

diagram as shown below:



What we've shown is that the tensor in Eq. (1.2.8) is equal to the tensor in Eq. (1.2.10). We can now imagine multiplying with $R(u+v)^{-1}$ to the right of Eq. (1.2.8) which gets rid of the purple vertex. Multiplying with $R(u+v)^{-1}$ to the right of Eq. (1.2.10) adds an extra vertex at the right end. If we now impose periodic boundary conditions along the horizontal legs and contract them, then $R(u+v)$ and $R(u+v)^{-1}$ cancel out by the cyclicity of the trace and we get

$$T(u)T(v) = T(v)T(u) \quad (1.2.11)$$

which is what we wanted to show!

1.2.4 Conservation laws

The Yang-Baxter equation, therefore, is the reason transfer matrices corresponding to different spectral parameter commute in the six-vertex model. You might wonder how does commuting transfer matrices help us? I'll not go into full details of how this works, but Baxter's book [8] has a very detailed discussion. An immediate consequence of commuting transfer matrices is that the operators which appear as co-efficients in the Laurent series expansion of $T(u)$ commute with each other. Suppose $T(u)$ has a Laurent series expansion of the form

$$T(u) = \sum_{n=-\infty}^{\infty} t_n u^n. \quad (1.2.12)$$

Since $T(u)T(v) = T(v)T(u)$, it follows that

$$[t_n, t_m] = 0. \quad (1.2.13)$$

There is an interesting consequence of this equation which I want to discuss before we move on. It turns out that t_1 above is actually the Hamiltonian of the XXZ Heisenberg model [9]. What we get therefore is an infinite set of conservation laws for the XXZ model. This is not a surprise because the two-dimensional classical six-vertex model is equivalent to the one-dimensional quantum Heisenberg XXZ model! Since we have an infinite set of conservation laws, it isn't hard to believe that the model is solvable.

There is another approach that Lieb [2, 10, 11] used in a series of papers to solve the six-vertex model. His method goes by the name of the Bethe ansatz [12] and is a very

powerful technique to solve statistical physics models. Unfortunately, I will not cover the Bethe ansatz in these notes. Our concerns lie elsewhere.

Hopefully this example has convinced you that the Yang-Baxter equation works like magic in establishing the integrability of a model by producing infinitely many conservation laws. In fact, I urge you to contrast the approaches in Chapters 8 and 9 of Baxter's book to solve the six-vertex model to understand how elegant the Yang-Baxter equation is.

That is all I have to say about the six-vertex model. The next example I want to discuss is that of Hamiltonian mechanics. Before that, however, I want to introduce a side character of our hero: *the quasiclassical r -matrix* [13].

1.2.5 Quasiclassical r -matrix

Typically, the R -matrix in the context of an integrable quantum model admits an expansion in $\hbar$ of the form:

$$R(u) = \mathbb{1} + \hbar r(u) + \mathcal{O}(\hbar^2). \quad (1.2.14)$$

The $r(u)$ is called the quasiclassical r -matrix. Since R satisfies the Yang-Baxter equation, $r(u)$ satisfies the *classical Yang-Baxter equation*:

$$[r_{12}(u), r_{13}(u+v)] + [r_{12}(u), r_{23}(v)] + [r_{13}(u+v), r_{23}(v)] = 0. \quad (1.2.15)$$

Exercise 1.3. *Show this.*

When we discuss classical mechanics, this is the version of the Yang-Baxter equation that appears. The classical r -matrix need not always exist. A famous example where it doesn't exist is the chiral Potts model [14, 15].

1.3 Hamiltonian Mechanics

The setup is as follows. We have a $2n$ -dimensional phase space Γ charted by canonical variables (q, p) along with a Hamiltonian function $H(q, p)$. Hamilton's equations of motion are

$$\dot{q} = \frac{\partial H}{\partial p}, \quad \dot{p} = -\frac{\partial H}{\partial q}. \quad (1.3.1)$$

I'm being sloppy and not writing down the indices explicitly. To understand how the classical Yang-Baxter equation appears in this setting, we first need to understand what makes this system of equations integrable in the first place. This notion is captured by the Liouville-Arnold theorem [16, 17].

Theorem 1.1 (Liouville-Arnold). *Suppose there exists n functions on phase space $f_1(q, p)$, $f_2(q, p)$, $\dots$, $f_n(q, p)$, such that*

1. *they include the Hamiltonian,*
2. *they mutually Poisson commute: $\{f_i, f_j\} = 0$ for all i and j , and*

3. they are functionally independent of each other, i.e., there is no relation of the form $\mathcal{F}(f_1(q, p), \dots, f_n(q, p)) = 0$.

Then Hamilton's equations of motion can be solved by quadratures. There exist canonical co-ordinates (Ω, ψ) such that Hamilton's equations are equivalent to

$$\dot{\Omega}_i = \psi_i, \quad \dot{\psi}_i = 0. \quad (1.3.2)$$

The variables (Ω, ψ) are called the action-angle variables.

A system that satisfies the assumptions of the above theorem is called *Liouville integrable*, and this is the definition of integrability we will adopt.

Exercise 1.4. Find a proof of the Liouville-Arnold theorem. As food for thought, can you come up with another sensible definition of integrability in Hamiltonian mechanics that is different from Liouville-Arnold integrability?

The Yang-Baxter equation is not transparent in this way of thinking about integrability. To bring out the Yang-Baxter equation, we will use the notion of a Lax pair [18, 19].

1.3.1 Lax Pairs

Suppose our Hamilton's equations of motion can be rewritten in the form:

$$\frac{dL}{dt} = [M, L] \quad (1.3.3)$$

where M and L are some $d \times d$ square matrices called the Lax pair.

For example, consider the Lax pair given by

$$L = \begin{pmatrix} p/m & \omega q \\ \omega q & -p/m \end{pmatrix}, \quad M = \begin{pmatrix} 0 & -\omega/2 \\ \omega/2 & 0 \end{pmatrix} \quad (1.3.4)$$

Then Eq. (1.3.3) with the above choice of L and M reproduces the familiar equation of motion of a harmonic oscillator.

If the equations of motion can be written in terms of a Lax pair, then there's a beautiful way of constructing conserved quantities. What we do is take traces of powers of L . The claim is that

$$Q_k = \text{Tr } L^k \quad (1.3.5)$$

are all conserved quantities thanks to Eq. (1.3.3). Since L is a $d \times d$ matrix, this gives us, at most, d conserved quantities. For the harmonic oscillator, all odd powers of L have zero trace while the even powers give powers of the Hamiltonian.

Suppose that the Lax pair gives us n functionally independent conserved quantities and also includes the Hamiltonian as part of the spectrum. How do we ensure that two such conserved quantities mutually Poisson commute, i.e.,

$$\{\text{Tr } L^r, \text{Tr } L^s\} = 0? \quad (1.3.6)$$

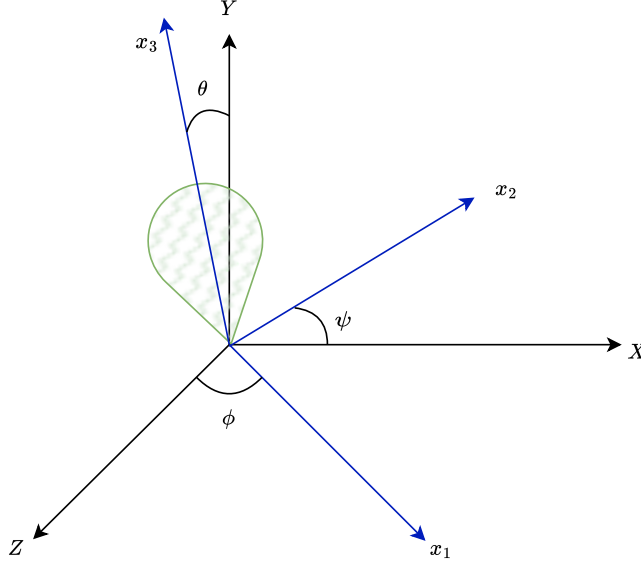


Figure 1.4: A sketch of the Euler top.

1.3.2 The Spectral parameter

Before addressing this question, however, I want to make a side remark about the spectral parameter which is best done with the example of the Euler top shown in Fig. 1.4. Recall that an Euler top's moment of inertia tensor is diagonal: $I = \text{diag}(I_1, I_2, I_3)$, and its angular momentum is $\vec{J} = I \cdot \vec{\omega}$. The equations of motion take the simple form

$$\frac{d\vec{J}}{dt} = -\vec{\omega} \times \vec{J} \quad (1.3.7)$$

which can easily be put in a Lax pair via the matrices

$$J_{ij} = \epsilon_{ijk} J_k, \quad \Omega_{ij} = \epsilon_{ijk} \omega_k. \quad (1.3.8)$$

In terms of these matrices, the equations of motion become

$$\frac{dJ}{dt} = [\Omega, J] \quad (1.3.9)$$

and we think of J and Ω as the Lax pair. From our discussion above, traces of powers of J give conservation laws. In this example, the Lax construction only gives us functions of $\vec{J}^2$: the Hamiltonian of the Euler top is not part of these conserved quantities.

From our discussion of the Liouville-Arnold theorem, it is necessary for the Hamiltonian to be part of the set of conserved quantities. To address this issue, we can introduce a spectral parameter $\lambda \in \mathbb{C}$ and consider a new Lax pair defined as

$$L(\lambda) = \mathcal{I}^2 + \frac{1}{\lambda} J, \quad M(\lambda) = \lambda \mathcal{I} + \Omega. \quad (1.3.10)$$

Here $\mathcal{I}$ is another diagonal matrix with entries $\mathcal{I}_1 = (I_2 + I_3 - I_1)/2$, etc. The equations of motion of the Euler top are equivalent to

$$\frac{d}{dt} L(\lambda) = [M(\lambda), L(\lambda)] \quad (1.3.11)$$

The Hamiltonian is now part of the spectrum of L , and we have also managed to introduce a spectral parameter.

Exercise 1.5. *Show that the Hamiltonian is now part of the spectrum of $L(\lambda)$. What is the lowest power of $L(\lambda)$ you need to consider to see the Hamiltonian? As you may be familiar, there are two other integrable tops that go by the name of the Lagrange and the Kowalewski top. Try writing their equations of motion using Lax pairs and calculate the conserved quantities coming from the Lax construction. Do these conserved quantities satisfy the assumptions of Liouville-Arnold integrability? You may find Babelon's book [20] useful for this exercise.*

1.3.3 Existence of the r -matrix

Let's now return to where we left off and ask how do we make two conserved quantities coming out of the Lax construction Poisson commute? With a spectral parameter in the game, we ask how to ensure

$$\{\text{Tr } L(\lambda)^r, \text{Tr } L(\mu)^s\} = 0 \quad (1.3.12)$$

for any $\lambda, \mu \in \mathbb{C}$ and any $r, s \in \mathbb{Z}$? The answer is rather simple. We first pull the two traces out of the Poisson bracket and write the above equation as

$$\text{Tr}_{12}\{L_1(\lambda)^r, L_2(\mu)^s\} = 0 \quad (1.3.13)$$

where $L_1 = \mathbb{1} \otimes L$ and $L_2 = L \otimes \mathbb{1}$ are matrices acting on two copies of the vector space on which L acts. A necessary and sufficient condition [20] for the above trace to be zero is the existence of an r -matrix such that

$$\{L_1(\lambda), L_2(\mu)\} = [r_{12}(\lambda - \mu), L_1(\lambda)] - [r_{21}(\mu - \lambda), L_2(\mu)]. \quad (1.3.14)$$

This equation needs some unpacking. Here r_{12} is some matrix acting on two copies of the underlying vector space. To be explicit, it has the following components:

$$r_{12} |ij\rangle = \sum_{k,l} (r_{12})_{ij}^{kl} |kl\rangle, \quad (1.3.15)$$

while the matrix r_{21} flips the role of 1 and 2, i.e.,

$$r_{21} |ij\rangle = \sum_{k,l} (r_{12})_{ji}^{lk} |kl\rangle. \quad (1.3.16)$$

An alternative definition is $r_{21} = \Pi r_{12} \Pi$ where Π is the permutation map that exchanges $1 \leftrightarrow 2$.

It is easy to see that the existence of the r -matrix guarantees that the two conserved quantities Poisson commute. The Poisson bracket $\{L_1(\lambda), L_2(\mu)\}$ in Eq. (1.3.14) also needs to be a consistent Poisson bracket. In particular, it has to satisfy the Jacobi identity:

$$\{\{L_1(z_1), L_2(z_2)\}, L_3(z_3)\} + \text{cyclic perms.} = 0. \quad (1.3.17)$$

As discussed in more detail in [3], if we assume that the r -matrix itself is not dynamical, then the above Jacobi identity simplifies to

$$[r_{12}(z_{12}), r_{13}(z_{13})] + [r_{12}(z_{12}), r_{23}(z_{23})] + [r_{13}(z_{13}), r_{23}(z_{23})] = 0; \quad (1.3.18)$$

the classical Yang-Baxter equation with the notation $z_{ij} = z_i - z_j$.

Exercise 1.6. *Derive this. What does the equation look like when r is dynamical?*

The upshot of the above discussion is as follows. Demanding that conserved quantities Poisson commute gives us the r -matrix. The Jacobi identity of this Poisson bracket gives us the classical Yang-Baxter equation.

That is pretty much all I wanted to say about Hamiltonian mechanics. The next example I want to discuss is $(1+1)$ -dimensional quantum field theories.

1.4 Two-dimensional QFTs

The third class of models where the appearance of the Yang-Baxter equation is more direct is $(1+1)$ -dimensional integrable QFTs. I will begin by explaining what definition of integrability we will adopt and how it forces us to consider only $(1+1)$ -dimensional theories. The main reference I'll follow is a beautiful paper by Witten and Shankar [21].

We will define integrability of a QFT via the following requirements:

1. There exist infinitely many conserved charges.
2. These charges come from integrals of local current densities, transforming under higher and higher representations of the Lorentz group. For example, there must be a charge $Q_{\mu\nu}$ that transforms as a rank-two tensor under the Lorentz group and associated to this charge is a current $J_{\mu\nu\rho}$ such that

$$Q_{\mu\nu} = \int_{\Sigma} J_{\mu\nu 0} dx^i \quad (1.4.1)$$

where Σ is any Cauchy slice.

3. These charges commute with the momentum P^μ and also among themselves. In particular, this means that these charges commute with the mass operator $M^2 = P^\mu P_\mu$.
4. The charges send one-particle states to one-particle states¹.

A famous theorem of Coleman and Mandula [22] says there are no QFTs satisfying the above requirements in spacetime dimensions greater than 2. We are therefore constrained to work in $(1+1)$ -dimensions if we want to talk of integrable QFTs in the above sense. I will discuss the precise statement of the Coleman-Mandula theorem now. But we will first need to review some definitions.

¹While I don't think this is a strict requirement, I'll throw it in just in case. I thank Omkar for pointing this out to me.

1.4.1 The Coleman-Mandula Theorem

The Hilbert space of any QFT is an infinite direct sum of multi-particle Hilbert spaces:

$$\mathcal{H} = \mathcal{H}^{(1)} \oplus \mathcal{H}^{(2)} \oplus \dots, \quad (1.4.2)$$

where $\mathcal{H}^{(n)}$ is the n -particle subspace. This n -particle subspace in turn is some subspace (due to the Pauli exclusion principle) of a tensor product of n single-particle subspaces. In terms of an equation,

$$\mathcal{H}^{(n)} \subset \mathcal{H}^{(1)} \underbrace{\otimes \dots \otimes}_{n \text{ times}} \mathcal{H}^{(1)}. \quad (1.4.3)$$

The S matrix of the theory is a unitary operator acting on $\mathcal{H}$.

We will call a unitary operator $U : \mathcal{H} \rightarrow \mathcal{H}$ a *symmetry of the S -matrix* if

1. it maps single-particle states to single-particle states,
2. it acts on multi-particle states as if they were tensor products of single-particle states,
3. it commutes with the S -matrix.

We will denote G to be the group of all symmetries of the S -matrix. Let G be a connected symmetry group satisfying the following five assumptions:

1. **Lorentz invariance.** The Poincaré group is a subgroup of G .
2. **Particle-finiteness.** For any finite $M > 0$, there are only finitely many types of particles with mass less than M . Recall that a particle type is an irreducible representation of P contained in $\mathcal{H}^{(1)}$.
3. **Weak elastic analyticity.** The scattering amplitudes are analytic functions of the scattering angles in some neighborhood of the physical region, except at points of normal incidence.
4. **Non-trivial scattering.** Let $|p, p'\rangle$ be a two-particle state and let T be the T -matrix defined as

$$S = 1 - i(2\pi)^{d+1} \delta(P_{out} - P_{in}) T. \quad (1.4.4)$$

We assume that

$$T |p, p'\rangle \neq 0 \quad (1.4.5)$$

except at special values of the Mandelstam variable s .

There is a final technical assumption which I won't discuss here that Coleman and Mandula assume. They think this assumption can be dropped with a more careful analysis, but they haven't managed to do it in their paper. Weinberg [23] shows that for a QFT, this assumption can be dropped.

If the group G satisfies all of these assumptions, then the group G is locally isomorphic to a direct product of an *internal symmetry group* and the Poincaré group. Stated differently, the Lie algebra of G is a direct sum of the Lie algebra of an internal symmetry

group and the Lie algebra of the Poincare group². An internal symmetry group is a group that commutes with the Poincaré group when considered as subgroups of G .

The proof is technical and you are welcome to take a look at [22] or [23]. As a consequence of this theorem, there are no integrable QFTs. However, in $(1+1)$ -dimensions, the analyticity assumption doesn't make sense because there are no scattering angles. Therefore integrable QFTs can (and do) exist in $(1+1)$ -dimensions.

1.4.2 Constrains on the $(1+1)$ -dimensional S -matrix

The presence of these higher charges places severe constraints on the $(1+1)$ -dimensional S -matrix. In fact, the S -matrix of an integrable $(1+1)$ -dimensional QFT satisfies the following constraints [4, 24, 25]:

1. **No particle production.** All $n \rightarrow m$ scattering amplitudes are zero if $n \neq m$.
2. **Elastic scattering.** The initial set of momenta equals the final set of momenta. Phrased differently, the final momenta is a permutation of the initial set of momenta.
3. **Factorizability.** Any $n \rightarrow n$ scattering amplitude is a product of $2 \rightarrow 2$ scattering amplitudes.

The proof of these facts goes as follows. Consider the action of a conserved rank-three charge $Q^{\mu\nu\alpha}$ on a single particle state $|p^a\rangle$ where a labels the spin and other internal quantum numbers of the particle. From Lorentz invariance and the assumption that the charge maps single-particle states to single-particle states, we have

$$Q^{\mu\nu\alpha} |p^a\rangle = M^{ab} p^\mu p^\nu p^\alpha |p^b\rangle \quad (1.4.6)$$

where M^{ab} is some matrix that dictates the transformation of the internal quantum numbers. We might as well set $M^{ab} = \delta^{ab}$ for all practical purposes by working in a basis where the action of Q is diagonal.

Since the charge $Q^{\mu\nu\alpha}$ is the integral of a local current density, its action on a well-separated many-particle state takes the form

$$Q^{\mu\nu\alpha} |p_1, \dots, p_n\rangle = \sum_{i=1}^n (p_i^\mu p_i^\nu p_i^\alpha) |p_1, \dots, p_n\rangle, \quad (1.4.7)$$

where I dropped the internal labels for notational convenience. The charge $Q^{\mu\nu\alpha}$ is conserved, which implies that if the incoming state $|p_1, \dots, p_n\rangle$ scatters off to $|k_1, \dots, k_m\rangle$, then we must have

$$\sum_{i=1}^n p_i^\mu p_i^\nu p_i^\alpha = \sum_{j=1}^m k_j^\mu k_j^\nu k_j^\alpha. \quad (1.4.8)$$

We can repeat the same argument with a rank-four charge, with a rank-five charge, and ad nauseam to get more and more constraints on the incoming and outgoing momenta. The only way to simultaneously satisfy these constraints is by requiring no particle production and elastic scattering. This proves the first two constraints on the S -matrix.

²Thanks to Abhirup for pointing this out to me.

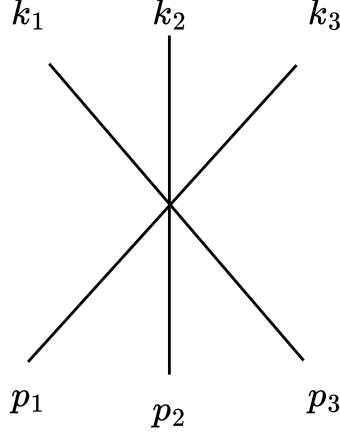


Figure 1.5:

To show factorizability, consider a $3 \rightarrow 3$ scattering process shown in Fig. 1.5. Now consider a rank n charge $Q^{\mu_1 \dots \mu_n}$ and consider the action of the operator

$$\mathcal{U}(c) = e^{icQ^{1\dots 1}} \quad (1.4.9)$$

where $Q^{1\dots 1}$ is the all-space component of $Q^{\mu_1 \dots \mu_n}$. From Eq. (1.4.7), the action of the above operator on a single-particle state is given by

$$\mathcal{U}(c) |p\rangle = e^{icp^n} |p\rangle. \quad (1.4.10)$$

The action of $\mathcal{U}(c)$ on a wavepacket centered at location $x = x_0$ and momentum $p = p_0$ is to shift

$$x_0 \rightarrow x_0 - ncp_0^{n-1} \quad (1.4.11)$$

as you can easily check. When $n = 1$, this is translating the center of the wavepacket by a constant amount. However, the rank- n charge Q shifts the center by an amount that depends on the momentum.

What the rank n charge Q lets us do to Fig. 1.5 is to shift the wavepackets relative to each other thereby obtaining the two scattering processes in Fig. 1.6. Since the charge

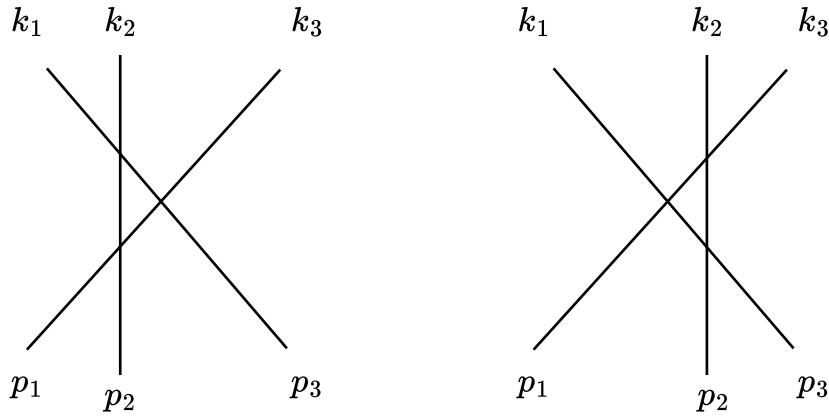


Figure 1.6:

Q is conserved, the operator $\mathcal{U}(c)$ does not affect elements of the S -matrix, therefore we require that the two scattering amplitudes in Fig. 1.6 and the scattering amplitude of Fig. 1.5 are all equal to each other.

Taking c to be very large will cause a large relative shift to the wavepackets which shows that the $3 \rightarrow 3$ scattering amplitude in Fig. 1.5 can be broken down into a product of three $2 \rightarrow 2$ scattering amplitudes. There are two different ways of breaking down the amplitude in Fig. 1.5 which are shown in Fig. 1.6, and for consistency we would require that the S -matrix satisfy the Yang-Baxter equation:

$$S_{12}(p_1 - p_2)S_{13}(p_1 - p_3)S_{23}(p_2 - p_3) = S_{23}(p_2 - p_3)S_{13}(p_1 - p_3)S_{12}(p_1 - p_2). \quad (1.4.12)$$

More generally, any $n \rightarrow n$ scattering amplitude can be factorized into a product of $2 \rightarrow 2$ amplitudes, and consistency of this factorization is given by the Yang-Baxter equation. The role of the R -matrix is played by the S -matrix of the QFT.

Chapter 2

Solutions to the Yang-Baxter Equation

In the previous chapter, we looked at three examples of physical systems: (a) the six-vertex model, which is an integrable statistical mechanics model, (b) Hamiltonian systems which are Liouville-Arnold integrable, and (c) $(1 + 1)$ -dimensional QFTs. The common thread connecting these three examples of integrable systems is the appearance of the Yang-Baxter equation in some form or the other.

In the six-vertex model, we saw that the Boltzmann weights could be parametrized in a funny way giving us a solution to the Yang-Baxter equation. In Liouville-Arnold integrable Hamiltonian systems, we saw that Poisson-commuting Lax charges gave us the classical r -matrix, and the Jacobi identity gave us the classical Yang-Baxter equation. In integrable $(1 + 1)$ -dimensional QFTs, we saw that our definition of integrability puts stringent conditions on the S -matrix. One of the conditions we derived was factorizability, and we showed that the S -matrix must satisfy the Yang-Baxter equation as a consistency condition.

As these examples suggest, the Yang-Baxter equation is ubiquitous in integrable models. While this is only a heuristic observation, we will adopt the point of view that all integrable models have the Yang-Baxter equation lurking in them. In this chapter, we will look for solutions to the Yang-Baxter equation with the hope of generating and classifying all integrable models.

I must apologize in advance because this chapter is going to be mathematical and dry. However, we will leave all this abstraction behind starting with the next chapter. The main reference I'll follow for the next section is a beautiful article by Jimbo [1], but I will try to fill in some of the details where necessary. Let's begin with solutions to the classical Yang-Baxter equation.

2.1 The Classical Yang-Baxter Equation

As we discussed in the previous chapter, the R -matrix usually admits an expansion in $\hbar$ of the form

$$R(u) = 1 + \hbar r(u) + \mathcal{O}(\hbar^2), \tag{2.1.1}$$

where we call $r(u)$ the quasiclassical r -matrix. The Yang-Baxter equation at $\mathcal{O}(\hbar^2)$ gives us the classical Yang-Baxter equation

$$[r_{12}(u), r_{13}(u+v)] + [r_{12}(u), r_{23}(v)] + [r_{13}(u+v), r_{23}(v)] = 0. \quad (2.1.2)$$

What do solutions of this equation look like?

Belavin and Drinfeld [26] proved a beautiful theorem classifying all solutions to the classical Yang-Baxter equation into three classes: (a) elliptic, (b) trigonometric, and (c) rational. In this section, I'll discuss their theorem.

The starting point for us will be the following observation. Recall that the R -matrix was a linear map from $V \otimes V \rightarrow V \otimes V$, where V was any finite-dimensional vector space. We can introduce some more mathematical structure to V by taking it to be a Lie algebra $\mathfrak{g}$ instead of a vector space V . We can then view the quasiclassical r -matrix as a map from $\mathfrak{g} \otimes \mathfrak{g} \rightarrow \mathfrak{g} \otimes \mathfrak{g}$ which can be expanded using the generators $\{X_a\}$ of $\mathfrak{g}$ as

$$r(u) = \sum_{a,b} r^{ab}(u) X_a \otimes X_b, \quad (2.1.3)$$

where each $r^{ab}(u) : \mathfrak{g} \otimes \mathfrak{g} \rightarrow \mathbb{C}$ is a function from $\mathfrak{g} \otimes \mathfrak{g}$ to $\mathbb{C}$, parametrized by the spectral parameter u .

Suppose we found some $r(u)$ for some Lie algebra $\mathfrak{g}$ that satisfies the classical Yang-Baxter equation. Here's one way to generate additional solutions. Pick any representation π of $\mathfrak{g}$, and look at the linear operator

$$(\pi \otimes \pi)(r(u)).$$

This operator too satisfies the classical Yang-Baxter equation! This is because representations preserve the structure of the Lie algebra. For example,

$$[r_{12}(u), r_{23}(v)] = \sum_{a,b,c,d} r^{ab}(u) r^{cd}(v) X_a \otimes [X_b, X_c] \otimes X_d, \quad \text{etc.} \quad (2.1.4)$$

Therefore, any $\mathfrak{g} \otimes \mathfrak{g}$ -valued solution to the classical Yang-Baxter equation is called a universal solution: it can be used to generate several other solutions.

Belavin and Drinfeld's theorem goes as follows. Suppose $\mathfrak{g}$ is a finite-dimensional simple Lie algebra with a $\mathfrak{g} \otimes \mathfrak{g}$ -valued solution $r(u)$ of the classical Yang-Baxter equation that is *non-degenerate*, i.e., $\det(r^{ab}(u)) \neq 0$. Suppose also that $r(u)$ is meromorphic in some neighborhood of $0 \in \mathbb{C}$. Then,

1. $r(u)$ extends to the whole complex plane with only simple poles.
2. The set of poles Γ of $r(u)$ is a discrete subgroup under addition in $\mathbb{C}$.
3. There are only three possible values the rank of Γ can take.

$$\text{rank}(\Gamma) = \begin{cases} 2, & \text{(elliptic)} \\ 1, & \text{(trigonometric)} \\ 0, & \text{(rational)}. \end{cases} \quad (2.1.5)$$

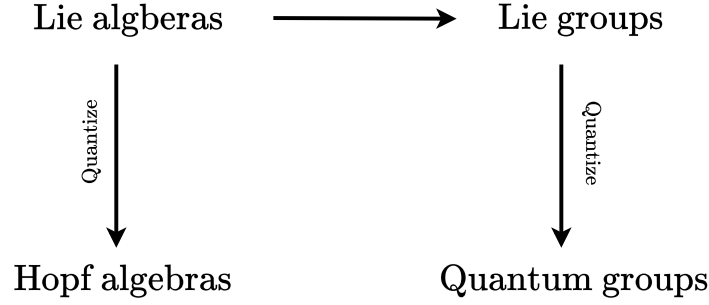


Figure 2.1:

Belavin and Drinfeld further showed that $\text{rank}(\Gamma) = 2$ only when $\mathfrak{g} = \mathfrak{sl}(n)$, and also that trigonometric and rational solutions exist for every $\mathfrak{g}$.

As an example, consider the rational solution

$$r(u) = \frac{1}{u} \sum_a X_a \otimes X_a. \quad (2.1.6)$$

The pole of $r(u)$ is at $u = 0$ which is a rank zero subgroup. An example of the trigonometric case is as follows. Let $\mathfrak{g} = \mathfrak{h} \oplus (\oplus_{\alpha} \mathfrak{g}_{\alpha})$ be the root space decomposition of the Lie algebra $\mathfrak{g}$. Let $X_{\alpha} \in \mathfrak{g}_{\alpha}$ be such that $(X_{\alpha}, X_{-\alpha}) = 1$. The function

$$r(u) = \sum_{\alpha > 0} (X_{\alpha} \otimes X_{-\alpha} - X_{-\alpha} \otimes X_{\alpha}) - \sum_a X_a \otimes X_a + \frac{2}{1 - e^u} \sum_a X_a \otimes X_a \quad (2.1.7)$$

is a trigonometric solution to the classical Yang-Baxter equation! The poles are on the imaginary axis and form an additive subgroup of rank 1 in $\mathbb{C}$.

We now turn to the question of finding solutions to the Yang-Baxter equation by *quantizing* these classical r -matrices. A natural question is where do these quantized r -matrices live? The classical r -matrix lives in a Lie algebra. It turns out that the right language to answer this question is that of quantum groups. I've tried to illustrate this connection in Fig. 2.1.

The main theorem we will prove is the following:

Theorem 2.1. *Let $(H, \mu, \eta, \Delta, \epsilon, R)$ be a braided Hopf bialgebra. Then the universal R -matrix satisfies*

$$R_{12}R_{13}R_{23} = R_{23}R_{13}R_{12} \quad (2.1.8)$$

the Yang-Baxter equation.

Understanding this theorem will require developing some mathematical machinery which will occupy us for the rest of this chapter. From here on, we will don our mathematician hats to make progress. The main references I will follow in the following sections are Drinfeld's ICM paper [27] and a paper by Chari and Pressley [28].

2.2 What is a quantum group?

Let me begin by briefly recalling what we mean by quantization. In classical mechanics, states are points on phase space, and observables are functions on phase space. Similarly, in quantum mechanics, states are rays in a Hilbert space and observables are linear operators on the Hilbert space. In both classical and quantum mechanics, the observables form an *algebra*. However, the algebra of classical observables is commutative while the algebra of quantum observables is not. Therefore, quantizing is like taking a commutative algebra and making it noncommutative.

In case you need reminding, an algebra A is a vector space over a field k with a *multiplication operation* from $A \times A \rightarrow A$ such that

$$(x + y) \cdot z = x \cdot z + y \cdot z, \quad z \cdot (x + y) = z \cdot x + z \cdot y, \quad (2.2.1)$$

and

$$(ax) \cdot (by) = (ab)(x \cdot y), \quad (2.2.2)$$

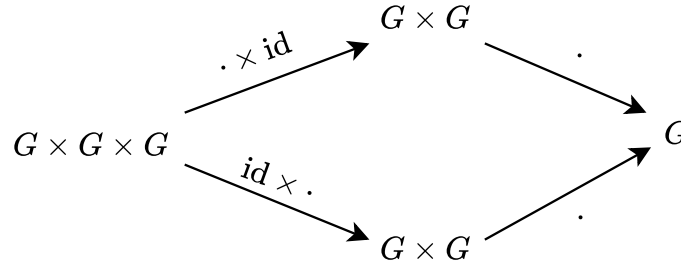
where $a, b \in k$, and $x, y, z \in A$. We will add to this definition a *unit* defined by the map $\eta_A : k \rightarrow A$ whose image is contained in the centre of A .

We will now play the same game with a group G . We will consider elements of G as states, and functions $\mathcal{F}$ on G as observables. The algebra of observables can be given the mathematical structure of a *Hopf algebra*. However, the Hopf algebra of functions of a group G is commutative. A quantum group is obtained by turning a commutative Hopf algebra into a noncommutative algebra.

Recall that a group G is a set endowed with a binary operation called multiplication $\cdot : G \times G \rightarrow G$ such that the following properties hold:

- **Associativity.** $g_1 \cdot (g_2 \cdot g_3) = (g_1 \cdot g_2) \cdot g_3$,
- **Existence of identity.** There exists an element $e \in G$ such that $g \cdot e = e \cdot g = g$,
- **Existence of inverses.** For every $g \in G$, there exists $g^{-1} \in G$ such that $g^{-1} \cdot g = g \cdot g^{-1} = e$.

These facts are summarized in the commutative diagrams below



(a) Associativity

$$\begin{array}{ccccc}
\{e\} \times G & \xrightarrow{\quad} & G \times G & \xleftarrow{\quad} & G \times \{e\} \\
\parallel & & \downarrow \cdot & & \parallel \\
G & = & G & = & G
\end{array}$$

(b) Existence of identity

$$\begin{array}{ccccccc}
G \times G & \xleftarrow{\text{id} \times \text{inv}} & G \times G & = & G & = & G \times G \xrightarrow{\text{inv} \times \text{id}} G \times G \\
& & \searrow \cdot & & & & \nearrow \cdot \\
& & & & \{e\} & &
\end{array}$$

(c) Existence of inverse.

If A is the set of all maps from G to some field (say $\mathbb{C}$), then there is a natural algebraic structure it comes with. In fact, A is a commutative associative algebra with a unit. However, we can give it further structure by thinking of A as being *dual* to G . The multiplication operation of the group G then induces an operation called *comultiplication* on A :

$$\Delta : A \rightarrow A \otimes A. \quad (2.2.3)$$

If $\mathfrak{F} \in A$ is a function $\mathfrak{F} : G \rightarrow \mathbb{C}$, then $\Delta(\mathfrak{F})$ acts on the tuple $(g, h) \in G \times G$ as

$$\Delta(\mathfrak{F})(g, h) = \mathfrak{F}(g \cdot h).$$

The associativity of the group multiplication induces *coassociativity* of the comultiplication denoted via the following commutative diagram:

$$\begin{array}{ccccc}
& & A \otimes A & & \\
& \nearrow \Delta & & \searrow \text{id} \otimes \Delta & \\
A & & & & A \otimes A \otimes A \\
& \searrow \Delta & & \nearrow \Delta \otimes \text{id} & \\
& & A \otimes A & &
\end{array}$$

Essentially, we reversed all arrows in the commutative diagram for associativity of the group G , replacing G with A and $\cdot$ with Δ .

Similarly, the existence of identity and inverses in the group G induces the existence of the *counit* $\epsilon : A \rightarrow \mathbb{C}$ and the *antipode* $S : A \rightarrow A$. They are defined by the following equations:

$$\epsilon(\mathfrak{F}) = \mathfrak{F}(e), \quad S(\mathfrak{F})(g) = \mathfrak{F}(g^{-1}), \quad (2.2.4)$$

where $\mathfrak{F} \in A$ is some element of A , e is the unit of G , and $g \in G$ is some arbitrary element of G . The counit and antipode are such that the following diagrams commute:

$$\begin{array}{ccccc} \mathbb{C} \otimes A & \xleftarrow{\epsilon \otimes \text{id}} & A \otimes A & \xrightarrow{\text{id} \otimes \epsilon} & A \otimes \mathbb{C} \\ & & \downarrow m & & \\ A & = & A & = & A \end{array}$$

$$\begin{array}{ccccccc} A & \xrightarrow{\Delta} & A \otimes A & \xrightarrow{\text{id} \otimes S} & A \otimes A & \xrightarrow{m} & A \\ & \searrow \epsilon & & & & \nearrow i & \\ & & \mathbb{C} & & & & \end{array}$$

$$\begin{array}{ccccccc} A & \xrightarrow{\Delta} & A \otimes A & \xrightarrow{S \otimes \text{id}} & A \otimes A & \xrightarrow{m} & A \\ & \searrow \epsilon & & & & \nearrow i & \\ & & \mathbb{C} & & & & \end{array}$$

In the above diagrams m is the usual multiplication operator in A defined as the product of the two functions evaluated at the same point in G . In other words, $m : A \otimes A \rightarrow A$ defined as $m(\mathfrak{F}, \mathfrak{G})(g) = \mathfrak{F}(g)\mathfrak{G}(g)$. The map $i : \mathbb{C} \rightarrow A$, called the *unit map*, is defined as $i(\lambda) = \lambda 1_A$ where 1_A is the identity in A . To be more explicit, $1_A : G \rightarrow \mathbb{C}$ defined as $1(g) = 1$.

The properties of $(A, m, \Delta, i, \epsilon, S)$ described above define a *commutative* Hopf algebra. More generally, any algebra with a comultiplication Δ , counit ϵ , and antipode S , is called a Hopf algebra. What we will do now is take the mathematical idea of a Hopf algebra and make contact with the Yang-Baxter equation. Before we get there, however, here are some examples of Hopf algebras.

2.2.1 Examples of Hopf Algebras

Group algebra of a finite group

The first example of a Hopf algebra I want to discuss is *dual*, in a sense I will make precise very soon, to the algebra of all functions $\mathfrak{F} : G \rightarrow \mathbb{C}$. Consider the algebra of all formal linear combinations of the form

$$\sum_{g \in G} c_g g$$

where $c_g \in \mathbb{C}$. Multiplication in this algebra is defined in the following natural way. A product of two such formal series is given by

$$\left(\sum_{g \in G} a_g g \right) \times \left(\sum_{h \in G} b_h h \right) = \sum_{g, h \in G} a_g b_h (g \cdot h).$$

This algebra can be given a Hopf algebra structure via the following definitions for the comultiplication, counit, and antipode, extended linearly:

$$\Delta(g) = g \otimes g, \quad \epsilon(g) = 1, \quad S(g) = g^{-1}. \quad (2.2.5)$$

This Hopf algebra is commutative iff G is commutative. A simple way to see that is by considering the elements g and h of the algebra where g and h are elements of the group. Their multiplication, by the above rule, is $g \cdot h$ which need not equal $h \cdot g$ if G is not commutative. For a non-commutative group G , therefore, this Hopf algebra is non-commutative. However, the dual of this algebra, which is the space of all functions from G to $\mathbb{C}$, is commutative. To show that these two algebras are dual to each other, consider any $\mathfrak{F} : G \rightarrow \mathbb{C}$ and any formal series $\sum_g c_g g$, and define their inner product as

$$\left\langle \mathfrak{F}, \sum_{g \in G} c_g g \right\rangle = \sum_{g \in G} c_g \mathfrak{F}(g). \quad (2.2.6)$$

The dual of a Hopf algebra is also a Hopf algebra, and in the two examples discussed above, either the Hopf algebra, or its dual, is commutative. These are not interesting algebras for our purposes. The more interesting Hopf algebras are those which are not commutative and whose duals are also not commutative. These enigmatic Hopf algebras are the ones that will give us universal solutions to the Yang-Baxter equation.

Universal enveloping algebra

The second example, which will play a key role soon, is the *universal enveloping algebra* $U\mathfrak{g}$ of a Lie algebra $\mathfrak{g}$. You can view this as a generalization of the previous example to Lie groups. The universal enveloping algebra is defined as the free algebra generated by the generators of $\mathfrak{g}$ subject to the commutation relations

$$\sum_c f_{ab}^c X_c = X_a X_b - X_b X_a, \quad (2.2.7)$$

where $\{X_a\}$ is the set of generators of $\mathfrak{g}$ and f_{ab}^c are the structure constants.

Recall that the free algebra on the alphabets $\{X_1, \dots, X_n\}$ is defined as the set of all words made up of strings of alphabets along with arbitrary linear combinations of such words. For example, on two alphabets X_1 and X_2 , the following are some valid elements of the free algebra:

$$X_1, \quad X_1X_2, \quad X_1^3X_2X_1, \quad \alpha X_1^3X_2X_1X_2^3 + \beta X_1X_2X_1^2,$$

where α and β come from $\mathbb{C}$.

The commutation relations that define the Lie algebra are to be modded out of the free algebra. In other words, two elements of the free algebra related by a commutation relation must be treated as equivalent, and the universal enveloping algebra is the set of all equivalence classes.

The universal enveloping algebra can be given the structure of a Hopf algebra via the following definitions of comultiplication, counit, and antipode:

$$\Delta(X_a) = X_a \otimes 1 + 1 \otimes X_a, \quad \epsilon(X_a) = 0, \quad S(X_a) = -X_a. \quad (2.2.8)$$

These definitions are to be extended to the full algebra such that they are algebra homomorphisms. For example, the comultiplication of a product of the form X_aX_b is defined as

$$\Delta(X_aX_b) = \Delta(X_a)\Delta(X_b) = X_aX_b \otimes 1 + X_a \otimes X_b + X_b \otimes X_a + 1 \otimes X_aX_b. \quad (2.2.9)$$

With these definitions, the universal enveloping algebra $U\mathfrak{g}$ turns into a *noncommutative* Hopf algebra when the Lie group is noncommutative.

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